C S Vineeth

Data Scientist

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EDUCATION

BMS Institute of Technology and Management Bengaluru, India

B.E., Artificial Intelligence and Machine Learning 2019 – 2023

EXPERIENCE

CBK Infotech India Pvt. Ltd. Bengaluru, India

Technical Consultant Apr 2023 – Present

Fraud Detection & Risk Intelligence

- Analyzed 500K+ transactional and relational records to design fraud-risk features and resolve severe class imbalance, lifting detection to ROC-AUC 0.9995 / F1 0.986 and improving relational fraud catch-rate 2.3% via graph-aggregation feature tuning.
- Ran comparative experiments across classical ML, tabular transformers, and GNNs to isolate which transactional and relational signals truly drove fraud, delivering risk scores and decision-support analysis for registry-grade transactions.

Cybersecurity & IoT Anomaly Detection

- Explored 211K+ network and IIoT flow records (44 features), using mutual information and SHAP to identify and validate the behavioral signals separating attacks from normal traffic, reaching AUROC > 0.999 across three model families.
- Modeled 2.2M+ time-series sensor readings with LSTM and Transformer autoencoders, engineering 221,976 sliding-window sequences and reconstruction-error thresholds to surface zero-day anomalies for monitoring.

NLP & Predictive Analytics

- Built NLP models for depression detection (XLNet, BiLSTM, CNN) fusing linguistic and psychological features, and analyzed 10M+ behavioral learning events to predict engagement and outcome-risk patterns.
- Addressed class imbalance and dataset bias through experimentation, label refinement, and benchmarking to improve generalization of text-classification models.

Medical Imaging & Computer Vision

- Trained and evaluated YOLO dental-detection models (Precision 0.9843, Recall 0.9820, mAP@50 0.9861), analyzing error patterns and embedding-based similarity for cross-modal biometric matching.
- Ran multi-dataset modeling studies across diabetic-retinopathy ordinal grading, osteoporosis risk (NHANES + imaging), EEG seizures, and gait analysis, focusing on data quality, label refinement, and metric-driven comparison.

TCR Innovation Mumbai, India

AI Research Intern Nov 2022 – Jan 2023

- Built computer-vision models for object counting, defect detection, and segmentation, improving accuracy 17% over baseline.
- Designed hybrid recommendation pipelines combining collaborative filtering, content-based similarity, and sentiment-derived features.
- Analyzed model performance and iterated on features and evaluation across multiple AI prototypes.

SKILLS

Languages & Data: Python, SQL, Pandas, NumPy

Analysis & Features: EDA, feature engineering, feature selection (mutual information, SHAP), class-imbalance handling

Machine Learning: scikit-learn, XGBoost, LightGBM, Random Forest, ensemble methods

Deep Learning: PyTorch, TensorFlow, CNNs, LSTM, Transformers

Evaluation & Impact: ROC-AUC, PR-AUC, F1, cross-validation, threshold tuning, model comparison & decision support